

Technology Stack

Date	2 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well Being
Maximum Marks	2 Marks

Define Your Technology Stack:

The table below identifies the programming languages, frameworks, databases, front-end tools, back-end tools, and deployment platform used to build the HDI Prediction System, along with a brief justification for each choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	React (with TypeScript) + Tailwind CSS, built with Vite	Provides a fast, component-based UI for the indicator input form and results view; TypeScript adds type safety, and Tailwind speeds up building a clean, responsive layout across devices.
2	Backend / Server-Side	Python (Flask)	Flask exposes a lightweight REST endpoint that receives the four input indicators, runs them through the trained prediction model, and returns the HDI score and development tier; Python is a natural fit alongside the ML/data-science tooling used to build the model.
3	Database / Data Storage	Not required at runtime — a static historical HDI dataset (CSV) was used only during model training	The prediction service is stateless: no user data or prediction history needs to be persisted, so no live database is part of the deployed architecture.
4	Cloud / Hosting / Deployment	Bolt.host	Bolt.host was used to build and deploy the application directly from the Bolt.new development environment, giving a quick, zero-configuration way to host and share the live prediction tool.
5	Version Control & CI/CD	GitHub	GitHub is used to track source code changes and collaborate on the project; it also integrates cleanly with the Bolt.new/Bolt.host deployment workflow for pushing updates.
6	Third-Party APIs / Other Tools	None required	All functionality (indicator input, prediction, and HDI tier classification) is self-contained within the

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
			application and its trained model, so no external third-party APIs are needed.